Analysis and Redesign of a Kenmore Smoothie Maker
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Product Description

A smoothie maker is a compact kitchen appliance that is used to quickly blend fruits, vegetables, liquids, and other foods into a smooth and consistent beverage. It features a durable plastic exterior that is easy to clean and a high-speed blade assembly inside the unit, which efficiently chops and mixes ingredients. Some models also include a serving cup for easy removal and service. The annual expected production volume is 15 million products. The components and CAD models are shown below in Figures 1 and 2.

Figure 1: Kenmore CAD Assembly

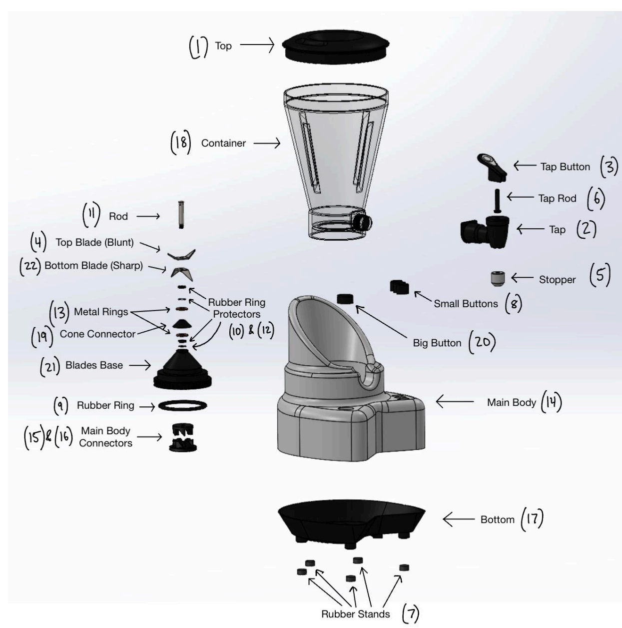


Figure 2: Kenmore CAD Exploded Assembly

Primary and Secondary Customers

The primary customer for the Kenmore product is people who make smoothies. This could apply to a variety of people, ranging from home chefs to the restaurant industry. The secondary customers within the primary customer group that were chosen are caterers who make smoothies and would bring the smoothie maker to a function or event.

Functional Analysis

As shown in the FAST diagram in Figure 3, the main function of the device is to make smoothies. This is accomplished primarily through stirring the mixture by chopping ingredients and providing power as well as by controlling operation by dispensing the mixture, containing the mixture, and reducing noise. Furthermore, there are additional functions like assuring dependability, assure convenience, and enhancing the product.

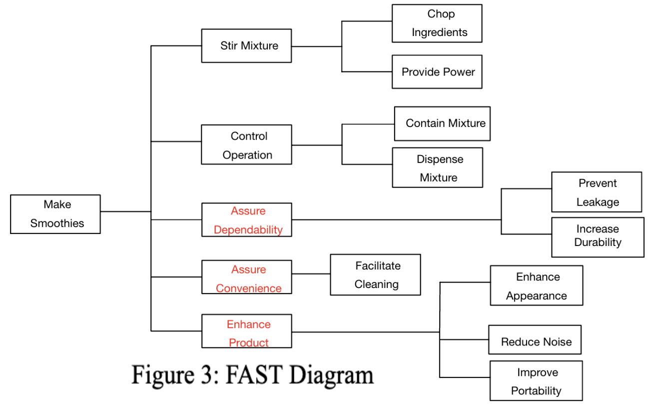


Figure 3: FAST Diagram

Function Component Mapping Matrix

The Function Component Mapping Matrix is shown in Table 1, which illustrates the relationship between the components of the smoothie maker and their corresponding functions. Most components serve two or more functions. The components related to the blades provide the power required to stir the mixture. The container is responsible for containing the mixture. The tap-related components are used to dispense the mixture. The buttons are also responsible for providing power. Besides these primary functions, most components also perform additional functions such as preventing leakage, increasing durability, and so on.

Table 1: Function Component Mapping Matrix

	Chop Ingredients	Provide Power	Contain Mixture	Prevent Leakage	Facilitate Cleaning	Dispense Mixture	Improve Portability	Increase Durability	Enhance Appearance	Reduce noise
Top			x	x			x		x	x
Tap				x		x				
Tap button						x				
Top blades	x	x						x		x
Stopper						x				
Tap rod						x				
Rubber stand							x	x		x
Small button		x							x	
Rubber ring				x						
Rubber ring protector				x				x		
Rod		x								
Rubber ring protector 2nd				x				x		
Metal ring				x				x		
Main body		x								x
Main body connector		x								
Connector		x								
Bottom							x	x	x	
Container			x	x	x	x	x		x	
Cone connector		x								
Big button		x							x	
Blades base		x								
Bottom blades	x	x						x		x

Function Differentiation Table

The function differentiation table, Table 2, shows how the different functions from the FAST table vary between the primary and secondary customer groups. The main functions that are different between customer groups include how operation is controlled and portability, which would be higher concerns for catering, as well as methods for dispensing/containing the mixture which is also a main concern.

Table 2: Function Differentiation Table

Same For Customer Groups	Different For Customer Groups
• Chop Ingredients • Provide Power • Reduce Noise • Prevent Leakage • Increase Durability • Facilitate Cleaning	• Contain Mixture • Enhance Appearance • Improve Portability • Dispense Mixture

Component Commonality Table

Table 3: Component Commonality Table

Platforms	Modules
• Top Blades • Bottom Blades • Blades Base • Rod • Rubber Stand • Main Body Connector • Connector • Cone Connector • Rubber Ring • Rubber Ring Protector • Rubber Ring Protector 2nd • Metal Ring • Bottom	• Tap Button • Small Button • Big Button • Tap Rod • Stopper • Tap • Container • Top

The component commonality table, Table 3, is based on the function-component mapping matrix and the function differentiation table. Platform components are used uniformly across all customer groups no matter the use of the smoothie maker. Module components are components that either can be swapped for a different part/assembly based on the use case, or could have a different function and maybe will not be used when a certain customer group operates the product. Some potentially controversial components were the container and tap which had multiple functions in both categories, they were chosen to be modular because while they are used for most operations, they serve different purposes based on intended use. On the flip side, the main body and bottom components had modular functions as well, but are platform parts because their use function is identical for all uses. Some of their functions change when other add-ons are added to the part base, but the individual part does not change functions. The components costs in the BOM was also taken into account. While most components' manufacturing costs are easily justified per their category, some notable components were the tap components with mid-low level unit costs that allow for customization without much profit loss, and the container with a high unit cost which can be easily redesigned to be cheaper, and could create profit opportunity with it being a modular part. The main body has the highest unit cost, but is necessary for the product to exist so it must stay a platform part with possibility for redesign. Base and connector parts have relatively low manufacturing costs which help offset the expensive main body in each final product.

CAD of Modules for Primary and Secondary Customers

To meet the functional requirements of the secondary customer group (professional caterers), specifically larger capacity, higher operational efficiency, and enhanced portability, three modules were modified from the primary household variant. The container volume was increased from 2.0L to 3.0L for larger batch preparation, and integrated side handles were added to the main body to improve portability. Additionally, the control buttons were color-coded for faster operator recognition while retaining identical geometry for interchangeability. Figure 4 presents a side-by-side CAD comparison of both product variants, with callouts highlighting these geometric and aesthetic modifications and linking them to their respective functional goals.

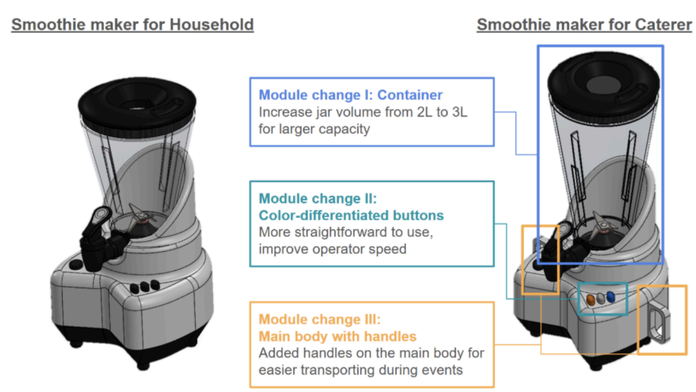


Figure 4: CAD Module Variants for Primary vs. Secondary Customers

Common Interfaces for Different Product Variants

The standardized interface architecture ensures seamless plug-in compatibility between the shared platform and all module variants, allowing components to be interchanged without redesign. Critical common interfaces, visualized in Figure 5, include the jar mount and drive coupling, which uses identical bayonet geometry, O-ring seating, and motor coupling to guarantee consistent mechanical alignment and sealing. Furthermore, the platform mounting interface shares the same base pattern, foot and boss arrangement, and datum faces, allowing both body designs to use the same structure. Finally, the control button interface maintains an identical housing geometry, preserving all electrical and mechanical connections and enabling the interchangeability of different button sets.

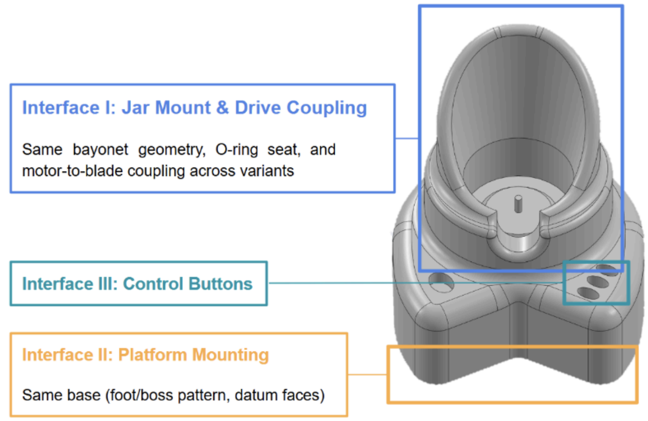


Figure 5: CAD of Common Interfaces

Product Family Assembly Diagrams

Taking the original assembly sequence that was identified through fishbone diagrams and Table 1, the inside out method of assembly shows the best ease of assembly and assembly features.

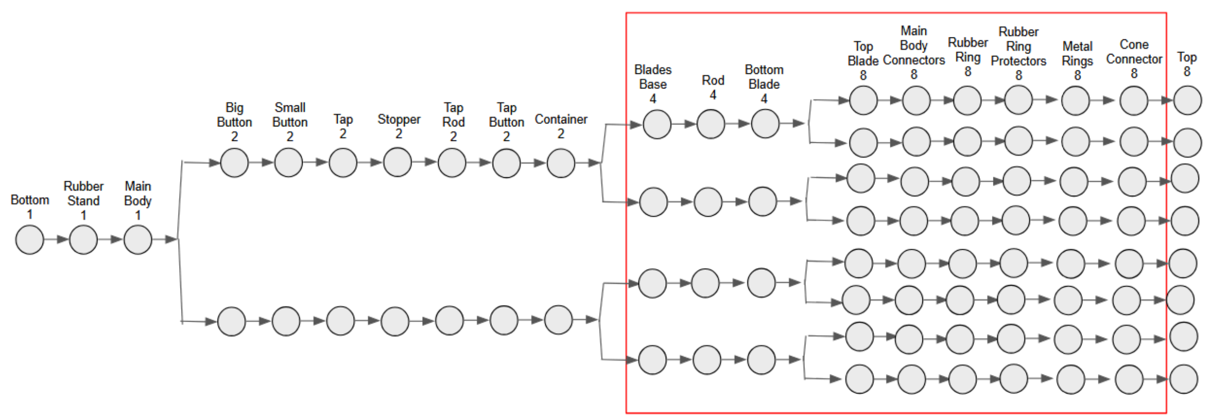


Figure 6: Original Product Family Assembly Sequence

We see in this assembly sequence the differentiation between the household user customer group and the catering group in the differing main body - with or without handles - and the blade functions - for heavy duty application, household blending, and household stirring. The blade differentiation is done with different combinations of the blades and a heavy duty version for longevity of the bottom blade. We also see the different container sizes, 2L and 3L for the different applications, however we do offer both sizes to both customer groups while we see that they may be favored one way or another. No delayed differentiation has been done for this model to minimize waste or reduce design complications.

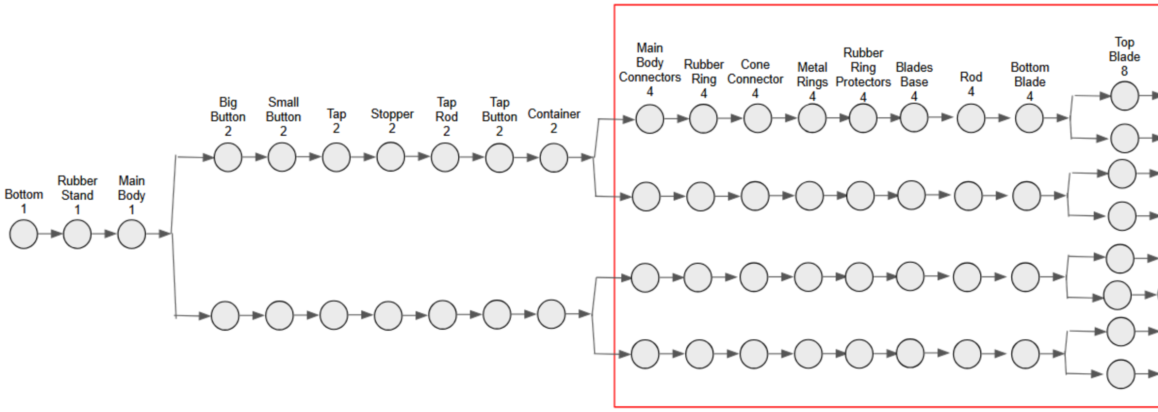


Figure 7: Revised Product Family Assembly Sequence (changes from original are highlighted)

Streamlining the assembly process, the rubber ring assembly was moved forward as they are shared in both product family assemblies. Similarly, the blades assembly process was moved backward to maintain the ease of assembly found earlier and reduce the number of unique assembly procedures needed. This means that the product differentiation for the module changes I, II, and III are now done at the end of the assembly process, reducing assembly line and supply chain loads.

Design for Manual Assembly

The manual assembly analysis of the primary product variant was performed, as shown in Table 4a. The total assembly time is 102.7 seconds and the total assembly efficiency is 55%.

Pre-DFA										
Item #	Name	# of Items	2-digit manual handling code	Manual handling time per part (sec)	2-digit manual insertion code	Manal insertion time per part (sec)	Total assembly time	Theoretical min # of parts	Assembly Efficiency (%)
1	Bottom	1	3 0	1.95	0 0	1.5	3.45	1	
2	Rubber Stand	4	0 1	1.43	3 0	2	13.72	1	
3	Main Body	1	3 0	1.95	0 0	1.5	3.45	1	
4	Big Button	1	1 0	1.5	3 0	2	3.5	1	
5	Small Buttons	3	2 0	1.8	3 0	2	11.4	1	
6	Tap	1	3 0	1.95	0 0	1.5	3.45	1	
7	Stopper	1	1 1	1.8	0 0	1.5	3.3	1	
8	Tap Rod	1	1 0	1.5	0 0	1.5	3	1	
9	Tap Button	1	3 0	1.95	0 0	1.5	3.45	1	
10	Container	1	3 0	1.95	0 0	1.5	3.45	1	
11	Bottom Connector	1	1 0	1.5	0 0	1.5	3	1	
12	Top Connector	1	3 0	1.95	0 2	2.5	4.45	1	
13	Rubber Ring	1	0 3	1.69	0 1	2.5	4.19	1	
14	Blades Base	1	1 0	1.5	0 0	1.5	3	1	
15	Cone Connector	1	1 0	1.5	0 0	1.5	3	1	
16	Metal Rings	3	0 3	1.69	0 0	1.5	9.57	0	
17	Rubber Ring Protectors	3	0 3	1.69	0 0	1.5	9.57	0	
18	Bottom Blade	1	2 3	2.36	0 0	1.5	3.86	1	
19	Top Blade	1	2 3	2.36	0 0	1.5	3.86	1	
20	Rod	1	1 0	1.5	0 0	1.5	3	1	
21	Top	1	1 0	1.5	0 0	1.5	3	1	
	Total	30					102.7	19	0.5552

Post-DFA										
Item #	Name	# of Items	2-digit manual handling code	Manual handling time per part (sec)	2-digit manual insertion code	Manal insertion time per part (sec)	Total assembly time	Theoretical min # of parts	Assembly Efficiency (%)
1	Bottom	1	3 0	1.95	0 0	1.5	3.45	1	
2	Rubber Stand	4	0 1	1.43	3 0	2	13.72	1	
3	Main Body	1	3 0	1.95	0 0	1.5	3.45	1	
4	Big Button	1	1 0	1.5	3 0	2	3.5	1	
5	Small Buttons	1	1 0	1.5	3 0	2	3.5	1	
6	Tap	1	3 0	1.95	0 0	1.5	3.45	1	
7	Stopper	1	1 1	1.8	0 0	1.5	3.3	1	
8	Tap Rod	1	1 0	1.5	0 0	1.5	3	1	
9	Tap Button	1	3 0	1.95	0 0	1.5	3.45	1	
10	Container	1	3 0	1.95	0 0	1.5	3.45	1	
11	Bottom Connector	1	1 0	1.5	0 0	1.5	3	1	
12	Top Connector	1	3 0	1.95	0 2	2.5	4.45	1	
13	Rubber Ring	1	0 3	1.69	0 1	2.5	4.19	1	
14	Blades Base	1	1 0	1.5	0 0	1.5	3	1	
15	Cone Connector	1	1 0	1.5	0 0	1.5	3	1	
16	Metal Rings	3	0 3	1.69	0 0	1.5	9.57	0	
17	Rubber Ring Protectors	3	0 3	1.69	0 0	1.5	9.57	0	
18	Blades	1	1 0	1.5	0 0	1.5	3	1	
20	Rod	1	1 0	1.5	0 0	1.5	3	1	
21	Top	1	1 0	1.5	0 0	1.5	3	1	
	Total	27					90.05	18	0.5997

(a) (b)

Table 4. Assembly analysis pre (a) and post (b) DFA

For design for manual assembly of a platform part we chose to work with the blades of the blender. These were chosen because of their high functional importance, as shown in the quad chart in Table 5, and the high assembly time from the assembly analysis performed above. For the module part we chose to work with the buttons on the machine, also because of their high functional importance and high assembly time.

Table 5: Quad Chart for Organizing based on functionality and manufacturing/assembly cost

		Manufacturing/Assembly Cost	
		Low	High
Functional Importance	Low	• Connector • Cone Connector • Tap Rod • Stopper • Metal Ring	• Tap Button • Tap
	High	• Rubber Stand • Rubber Ring • Rubber Ring Protector • Rubber Ring Protector 2nd • Main Body Connector • Rod • Big Button • Small Button	• Top • Top Blade • Bottom Blade • Blades Base • Main Body • Container • Bottom

With the aim of decreasing manufacturing time, the modifications done to the blades were combining both the top and bottom blade into one part, changing the inner hole to a square shape, and sharpening both blades, shown in Figure 8. These changes follow the following DFA guidelines: DFA-1: Reduce Part Count, DFA-6: Increase Part Symmetry. The modifications done to the buttons were to combine all 3 separate buttons into one part, filet both edges, and round the edges, Shown in Figure 8. These changes follow the following DFA guidelines: DFA-1: Reduce Part Count, DFA-6: Increase Part Symmetry.

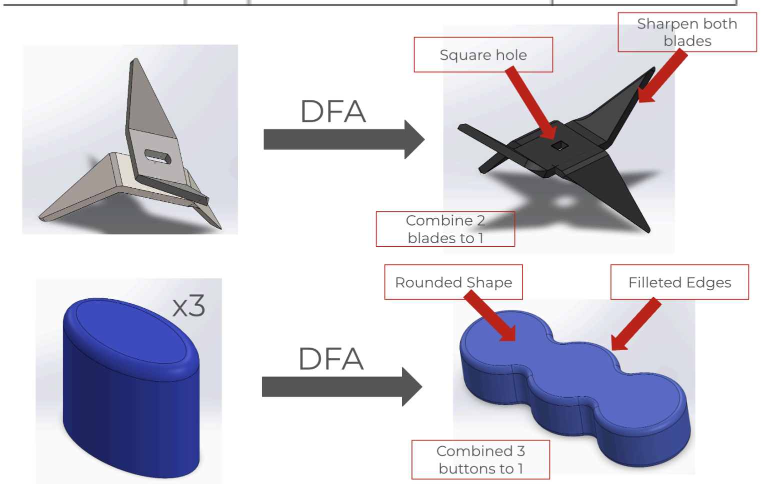


Figure 8: DFA Modifications

The assembly analysis was performed again with the new modifications to the small buttons and blades in mind. These modifications decreased the total assembly time from 102.7 seconds to 90.05 seconds and increased the efficiency from 55% to almost 60%, shown in Table 4b. These modifications should not have an effect on the functionality of the machine. These modifications will decrease manufacturing cost because they reduced part numbers and increased assembly efficiency.

Design for Automated Assembly

The subassembly chosen for analysis is the bottom connector to blades base subassembly. This was chosen because overall it has high importance to the design and many parts, leading to a high manufacturing time. The initial analysis was performed for the entire subassembly and the overall assembly cost for this subassembly was $69.80 with an efficiency of 21%, shown in Table 6.

Table 6: Pre-DFA Assembly Analysis

Item #	Name	# of Items	5-digit automatic handling code	2-digit automatic insertion code	Total assembly cost	Theoretical min # of parts	Assembly Efficiency (%)
11	Bottom Connector	1	05000	00	3.6	1	
12	Top Connector	1	05600	00	8.4	1	
13	Rubber Ring	1	05000	01	3.8	1	
14	Blades Base	1	15000	00	20.4	1	
15	Cone Connector	1	05000	00	4.8	1	
16	Metal Rings	3	05000	20	28.8	0	
	Total	8			69.8	5	0.2149

Based on the previous analysis, there are a few changes that can be made to the assembly to reduce assembly cost. First, we combined the top connector, cone connector, and blades base into one part. This allowed us to eliminate the rings used to attach these parts. Finally, we tapped the hole in the top of the blades base to allow for the same connection as before. These changes were made following the DFA guidelines, specifically: DFA-1: Reduce Part Count, DFA-2: Reduce Fasteners, DFA-10: Add Features for Easy Insertion, and DFA-12: Minimize turning over.

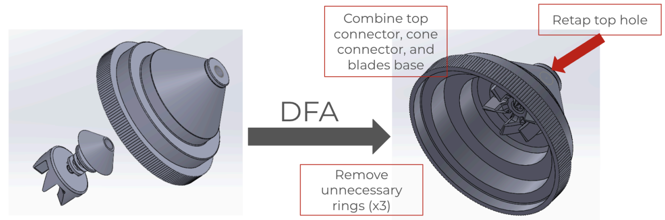


Figure 9: DFA Modifications

After these modifications, the assembly analysis was performed again. This resulted in the cost of assembly decreasing from $69.80 to $15.40 and the assembly efficiency increased from 21% to 97%, shown in Table 7. These modifications should not affect the functionality of the machine but they may increase manufacturing cost because of the more complex part design.

Table 7: Post-DFA Assembly Analysis

Item #	Name	# of Items	5-digit automatic handling code	2-digit automatic insertion code	Total assembly cost	Theoretical min # of parts	Assembly Efficiency (%)
11	Bottom Connector	1	05000	00	3.6	1	
12	New Combined Part	1	15600	00	8	1	
13	Rubber Ring	1	05000	01	3.8	1	
	Total	3			15.4	5	0.974026

Process Selection

For the button, a silicone rubber, an elastomer, was chosen for the ergonomic and tactile feel for the user. The shape is assumed to be a solid form. The critical dimension for this part is the length, 42.74mm, shown in the figure above. The function component mapping matrix indicates the button is used to enhance appearance and provide power to operate the equipment by the user. The length creates the visibility of the part and provides the tactile point for the user to engage with the on-off function of the equipment. Additionally, the thickness of the part is critical as it allows the user to input the button and trigger the internal switch, otherwise the distance the button can be pressed in.

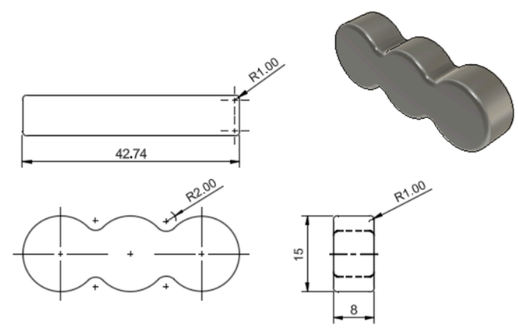


Figure 10: Button redesign part drawing with dimensions

Product Stats:Silicone Rubber (Elastomer), 32.784g, 15 million units, 42.741mm x 8mm, tolerance .10mm and surface roughness 1um, no secondary finishing						
Processes compatible with material	**Shape**	**Mass**	**Section Thickness**	**Tolerance**	**Surface roughness**	**Economic Batch Size**
Machining						Fail
Injection Moulding						
Compression Moulding		Fail				
Rotational Moulding	Fail					
Thermo-forming	Fail					
Polymer Casting		Fail				
Additive Manufacturing						Fail

Table 8: Process Selection Table for button redesign

Given the information found in the process selection table above, the button should be manufactured using an injection moulding process to reach the desired physical specifications and the production volume. No secondary finishing processes are necessary as the required tolerances and surface roughness can be achieved via the primary process. The tolerance for this part is 0.10mm as it needs to fit precisely with clearance in the cutout in the main body to function with minor flexibility due to the material itself.

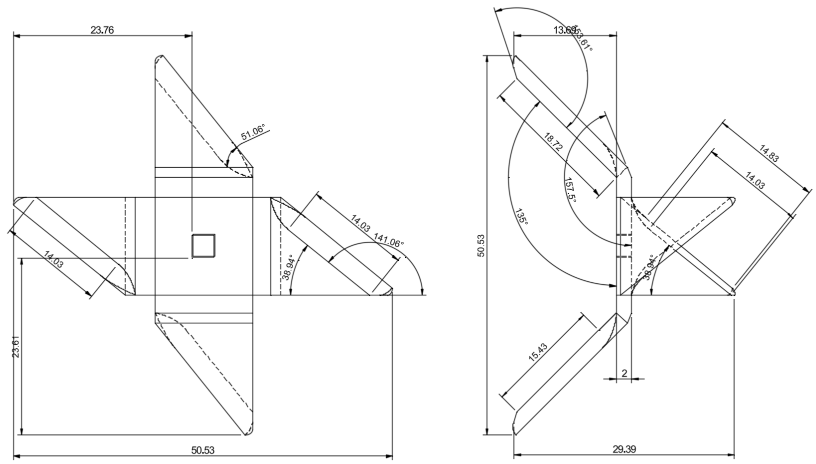


Figure 11: Blade redesign part drawing with dimensions

The critical dimension of the blade component is the thickness, 2mm, of the blade as its primary operation from the functional component mapping matrix is to chop ingredients with other functions of reducing noise and increasing durability. The thickness of the blade, and thus the sharpness, will determine the precision and effectiveness that the blade cuts through ingredients. With a sharper and more effective blade comes a quieter product. Additionally, the critical dimension of thickness determines the durability of the piece and ultimately the unit - with a thinner blade that may be prone to deforming comes increased sharpening and ultimately replacement costs.

Product Stats: Stainless Steel, 15.493g, 15 million units, 50.53mm length 2mm thickness, Tolerance .5mm and surface roughness 1um, secondary finishing						
Processes compatible with material	**Shape**	**Mass**	**Section Thickness**	**Tolerance**	**Surface roughness**	**Economic Batch Size**
Sand Casting	Fail					
Die Casting	Fail					
Investment Casting	Fail					
Centrifugal Casting	Fail					
Forging	Fail					
Extrusion	Fail					
Sheet Forming						
Powder methods	Fail					
Electro-machining					Fail	
Machining						Fail
Additive Manufacturing				Fail		
Finishing						
Precision Machining						
Grinding						
Lapping						
Polishing						

Table 9: Process selection table for the redesigned blade component.

The blade is made of stainless steel for corrosion resistance, food safety, and durability. This is a flat or folded sheet shaped by DFM Process Selection criteria. While tighter tolerances may be desired, to reach the other design specifications and economic batch size, the tolerances had to be lessened to choose a feasible operation. Ultimately, a tolerance of 0.5mm was deemed necessary at the attachment point, a washer can accommodate any flexibility within this tolerance. With this, the blades should be manufactured using a sheet forming process. Any of the secondary finishing operations are adequate and the most cost effective one should be chosen.

The cone and connector subassembly is to be made of Acrylonitrile Butadiene Styrene (ABS) as a strong, lightweight, and durable thermoplastic, adhering to food safety guidelines and for its rigidity. This is considered a solid 3D shape by DFM Process Selection criteria.

Design For Manufacturing - Plastic

We selected injection molding as the primary manufacturing process for the cone connector. Given an annual production volume of 15 million parts, the initial manufacturing cost was estimated at \$2.18 per part. Operator costs constituted the majority of the expense, primarily due to a maximum wall thickness of 14.08 mm, which resulted in a prolonged cooling time.

Table 10: Pre-DFM estimated cost per part

Cost of Injection Molding per Part	
Material cost	\$ 0.21
Operator cost	\$ 1.97
Mold cost	\$ 1.47×10^{-3}
Total Cost per Part	**\$ 2.18**

We utilized DFMPro to analyze the part geometry, highlighting issues regarding thick walls and insufficient draft angles. Following DFMPro and course guidelines, we redesigned the part by reducing the wall thickness to 3 mm (uniform) and adding 1-degree draft angles to facilitate ejection. As shown in Figure 12, these changes eliminate sink marks and ensure structural stability.

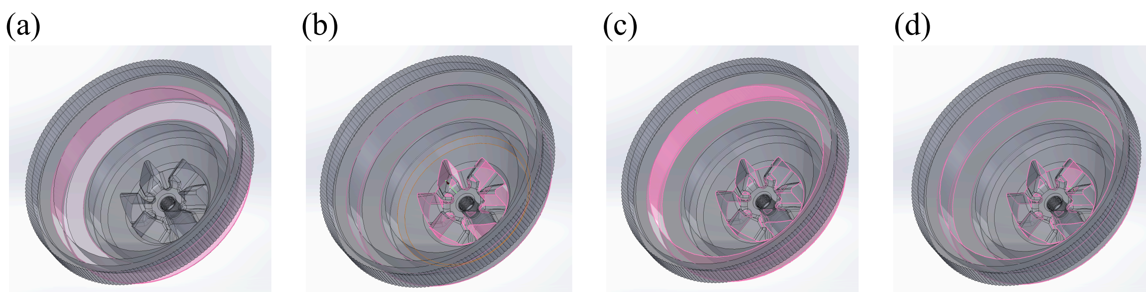


Figure 12. Pre-DFM cone connector design with DFMPro analysis (a) minimum radius at base of boss (b) minimum draft angle (c) uniform wall thickness (d) mold wall thickness

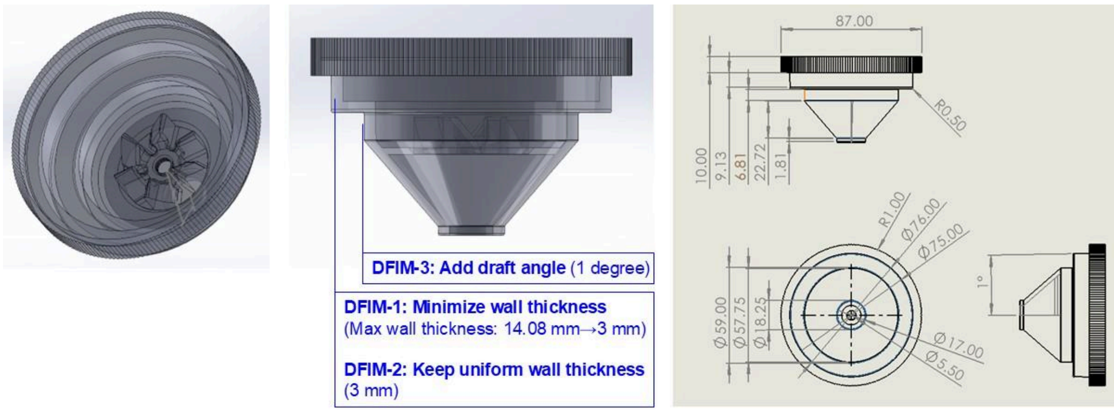


Figure 13. Post-DFM cone connector design, isometric view CAD

Table 11: Post-DFM estimated cost per part

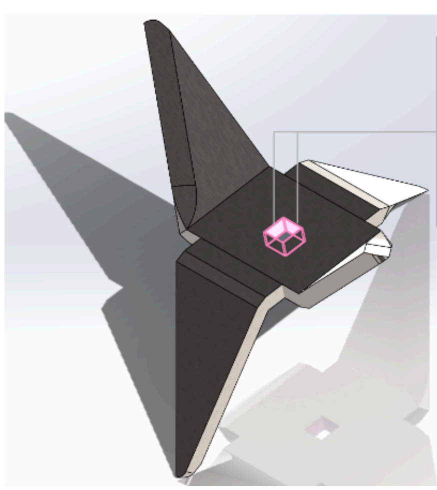
Cost of Injection Molding per Part	
Material cost	$ 0.21
Operator cost	$ 0.13
Mold cost	$ $1.47x10^{-3}$
Total Cost per Part	**$ 0.34**

The redesign drastically reduced cooling time. Assuming a 1-year payback period, the cost dropped to $0.34 per part, achieving total savings of $1.84 (~84%). Ultimately, the new design proves to be significantly more cost-effective while maintaining all necessary performance requirements.

Design For Manufacturing - Metal

In this part, we are analyzing the blade component, made from AISI 304 stainless steel (2 mm thickness) and manufactured via sheet metalworking. DFMPro identified an insufficient slot width-to-thickness ratio, corresponding to Design Guideline #7: Avoid narrow features. Manufacturing requires blanking, bending, and piercing dies. For a production volume of 15 million units per year, the estimated manufacturing cost is $0.4017 per part.

Post DFA, Pre DFM



DFMPro recommended increasing the "slot width to thickness ratio" to 2 or more (Use Slot Width >= 4.0 mm) (Currently ratio is 3 mm / 2 mm = 1.5)

DFSM-7: avoid narrow features

Cost of Sheet Metalworking per part	
Material Cost	$ 0.2380
Operating Cost	$ 0.1634
Die Cost*	$ 0.0003
Total cost per part	**$ 0.4017**

Instance Details

Slot Width (W) = 3.0 mm
Sheet Thickness (t) = 2.000 mm

Slot Width to Thickness Ratio (W/t) :
Actual Value = 1.5
Recommended Value >= 2.0 (Use Slot Width (W) >= 4.0 mm)

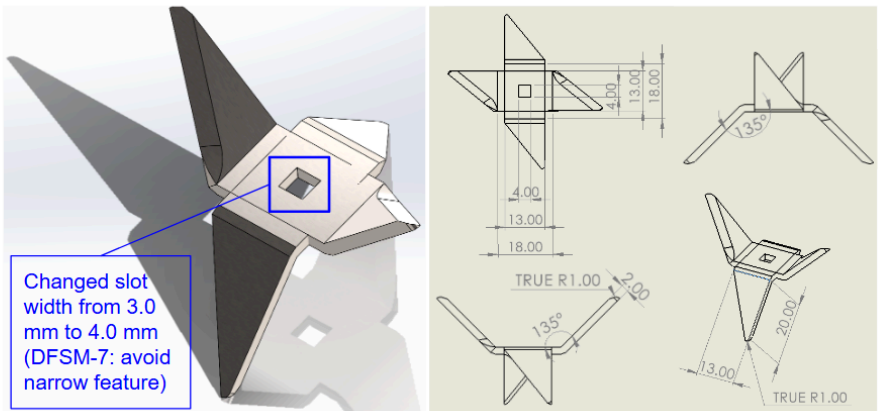
Rule Description

If the slot width falls below a minimum value, compressive stress on the punch increases dramatically, causing the thin and weak punch to buckle or break. Hence the slot width should increase with increasing slot length and sheet thickness.

Figure 14. Pre-DFM Blade Design, DFMPro Suggestions, Estimated Cost Per Part

In the analytical cost model for sheet metalworking, the redesigned part shows a slight cost increase, from $ 0.4017 to $ 0.4019 per unit. The bigger slot (from 9 mm² to 16 mm² in area) increases the shearing length and blank area, raising operating costs. Functionally, the redesign has no negative impact: the slot still meets clearance requirements, improves reliability by reducing interference risk, and does not compromise structural integrity.

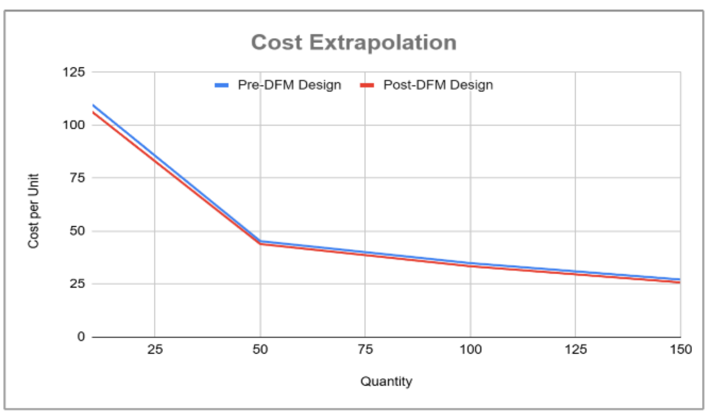
Post DFA and DFM



Cost of Sheet Metalworking per part	
Material Cost	$ 0.2380
Operating Cost	$ 0.1636
Die Cost*	$ 0.0003
Total cost per part	**$ 0.4019**

Figure 15. Post-DFM Blade Design, Isometric View CAD, Estimated Cost Per Part

Hubs provides CNC machining quotes because sheet metalworking is unavailable. Under CNC machining, wider slots simplify toolpaths and allow larger tools, reducing machining time and tooling costs. Thus, the redesigned part receives a lower CNC quote. The opposite cost trends compared to the analytical sheet metalworking estimates are expected, as the two processes have fundamentally different cost drivers.



	Analytical Method*	Hubs**
Pre-DFM Design	$ 0.4017	$ 54.31
Post-DFM Design	**$ 0.4019 (higher)**	**$ 52.41 (lower)**

*Analytical method used **sheet metalworking process**
Hubs quote used **machining process due to lack of quote availability for sheet metalworking

Figure 16. Extrapolation Excel Plot, Analytical vs. Hub Cost Estimation Comparison Table

Design of Product Service System

To further develop the product variants, a product service system was developed with caterers and home cooks in mind allowing customers to rent smoothie makers for events or holidays. Essentially the service system allows customers to rent their own smoothie maker using an online app or website, accessible by personal devices, with options for pickup and drop-off shown below in the conceptual design flow in Figure 17.

Join	Reserve	Obtain Product	Use Product	Return Product	Pay
Register	Mobile App	Deliverable with Fee	Optional Help Offered for Catering Events	Different Location	Automatic Billing
Pre-Register	Website	Distributed Locations		Same as Pickup	Check-out on Mobile Device
Company Bulk Registration	Return Time Designation	Contactless Pickup		Contactless Drop-off	Kiosk at Drop-off Location

Figure 17: Conceptual Design Flow

This product service system would allow for an improved customer service experience compared to the original design by allowing users to experience the product without needing to own it. Allowing customers to rent the smoothie maker for their favorite gatherings would not only result in increased customer satisfaction, but also could draw in more business, allow customer feedback, and reduce environmental impact all while being a cheaper option than buying the product. With customers renting out the smoothie maker, less waste would accumulate from unnecessary purchases and the team would be able to troubleshoot issues happening with the product, further developing the design and learning valuable information about failure modes.

Conclusions and Design Critiques

Altogether, the redesign of the Kenmore Smoothie Maker was highly successful. Through functional decomposition, design for manufacturing and assembly, and process optimization, the team achieved substantial improvements in both assembly and production efficiency. The revised design reduced assembly cost by $54.40 per part, reduced assembly time by $12.65 seconds, increased assembly efficiency by nearly 5%, and lowered manufacturing costs by $1.84 per part during the manufacturing process.

Overall, the new design is manufactured much easier and cheaper with fewer components, at the expense of more geometrically complex single parts, which would increase tooling costs. If the team could re-analyze the product design processes again, we would attempt to create a more modular design that could be assembled and reassembled by customers whenever necessary, as well as further optimizing more components for manufacturing and assembly, and making sure that the product design is inclusive for people with disabilities who may want to use the product.